Predicting the efficacy of malaria vaccines across the life-course

Supplementary Notebook 7. Estimates of severe malaria associated mortality

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Severe malaria and mortality data from Reyburn et al. (2005) was downloaded from the journal. The spreadsheet contains 1989 severe malaria episodes from hospital admissions from northeastern Tanzania collected over either six months (one hospital) or one year (nine hospitals) in 2002. Transmission intensity was stratified by village height above sea-level: high intensity below 600m, medium intensity from 600-1200m and low intensity above 1200m. There were 134 deaths in total.

The age distributions of severe malaria episodes and deaths, across transmission intensities, are shown in the Figure 1. The number of severe malaria episodes and deaths mirrors transmission intensity. As expected most cases occur in children less than 5 years of age, and the median ages of severe malaria and deaths reduce as transmission intensity increases (Table 1).

Transmission	Median age of severe malaria (years)	Median age of death (years)
high	1.1	1.0
medium	3.0	5.7
low	6.5	25.0

Table 1: Median ages of severe malaria and deaths across transmission intensities

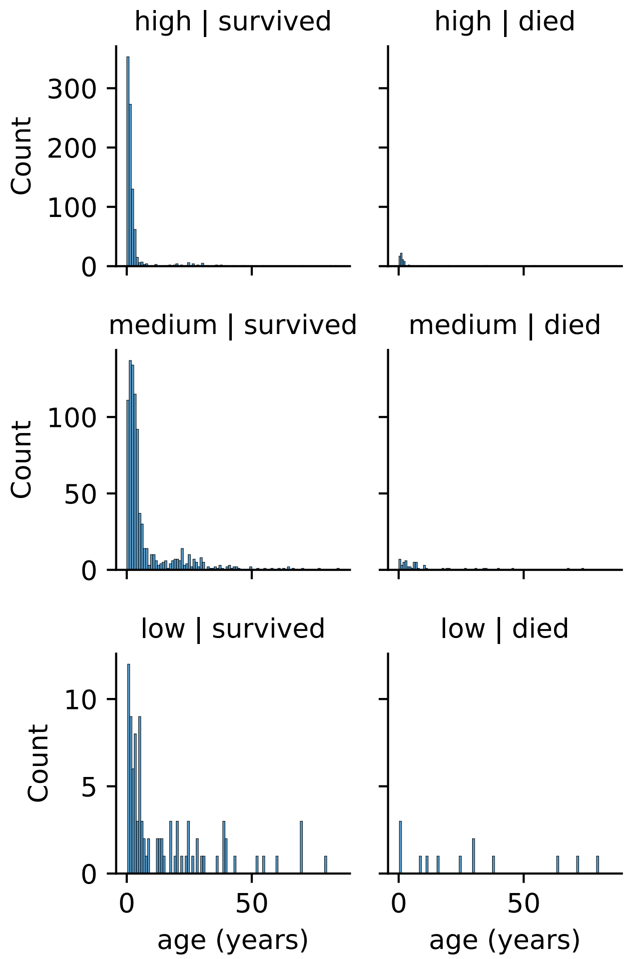

Figure 1: Raw counts of severe malaria episodes and deaths by age and transmission intensity from Reyburn et al. (2005).

1 Age-specific risk of severe malaria associated death

To visualise severe malaria associated age-specific death rates (Figure 2), the data are pooled across transmission intensities and binned with age boundaries of 0, 1, 2, 3, 4, 5, 10, 15, 20 and 80. Death rate is defined as the number of deaths in a bin divided by the number of severe malaria episodes in a bin.

Death rates show a decline from ages 0 to about 3-4 years, then show a rise at about 5 years and then a plateau thereafter. Above 20 years old, the scarcity of deaths makes the estimation unreliable.

The dip in death rate is corroborated by data from Marsh and Snow (1999), although the minimum occurs in 1 year olds, and Baird et al. (1998), that shows a dip in the 2-5 year olds, although the number of deaths is small.

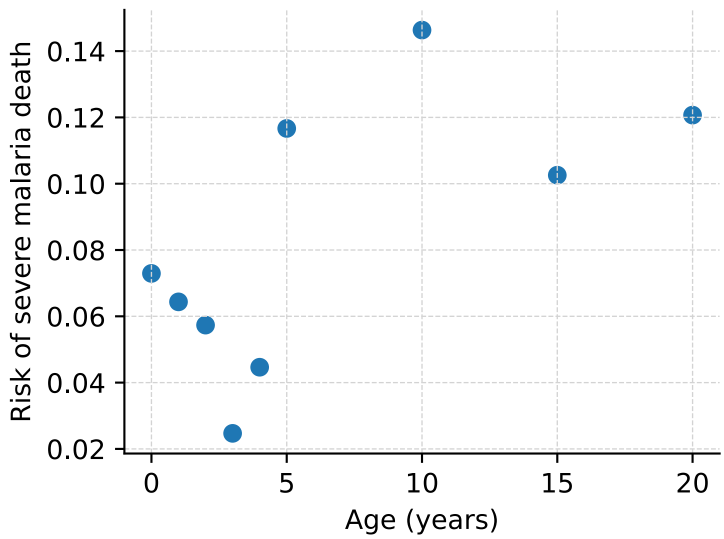

Figure 2: Age-specific risk of death among severe malaria cases pooled across transmission intensities from Reyburn et al. (2005).

2 Fit of GAM model to the raw data

To estimate the age-specific death rate we fitted a logistic generalised additive model with binomial errors to the raw data (up to 20 years old) using pyGAM (Servén and Brummitt 2018). We fit a spline term to age and a feature term to transmission intensity. The code is in Scripts/reyburn.py.

The partial dependence functions with 95% CIs (red dashed lines) of the two model terms (Figure 3) reveal a significant effect of age but not of transmission intensity. In other words, there is insufficient evidence to detect age-specfic death rates across transmission intensities.

The GAM was refit with transmission intensity removed and the estimated age-specific death rate used for our model.

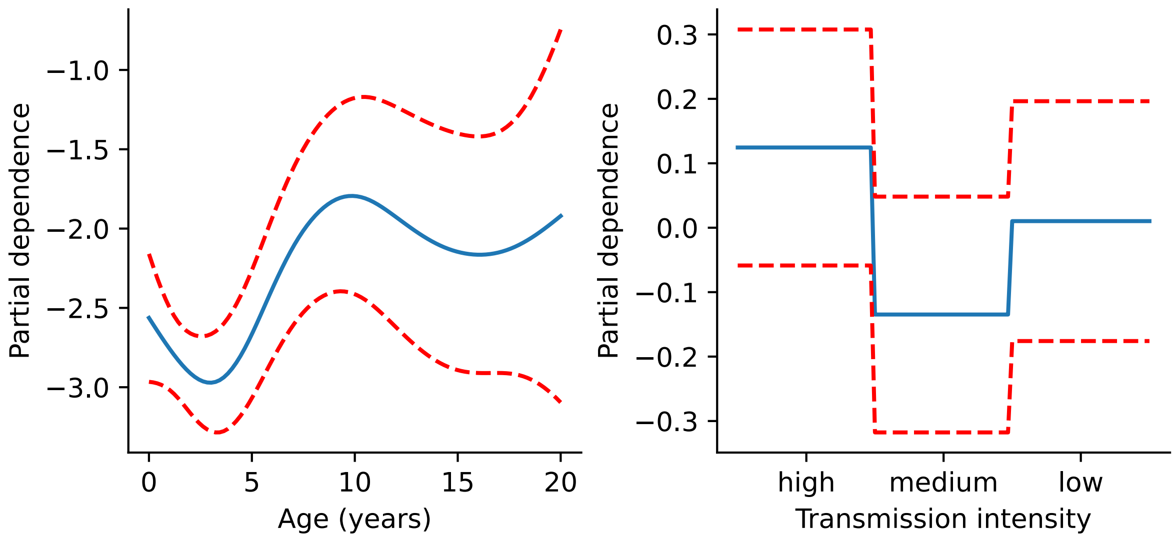

Figure 3: Partial dependence plots from the fitted GAM showing the effect of age and transmission intensity on the risk of death among severe malaria cases.

2.1 Estimated age-specific death rates

Above 10 years of age the estimate becomes highly uncertain due to sparsity of deaths. The drop in the curve in Figure 3 after its peak at 10 years old is an artifact of fitting GAM splines. Moreover, it is inconsistent with observations that adults are more at risk of death from severe malaria than children (Greenberg and Lobel 1990; Baird et al. 1998).

We therefore assume that the risk of death plateaus from age 10 onwards.

The estimated mean age-specific death rates (dashed line) with 95% CIs and the age-stratified data (circles) are shown in the Figure 4. Our model uses the mean age-specific death rates.

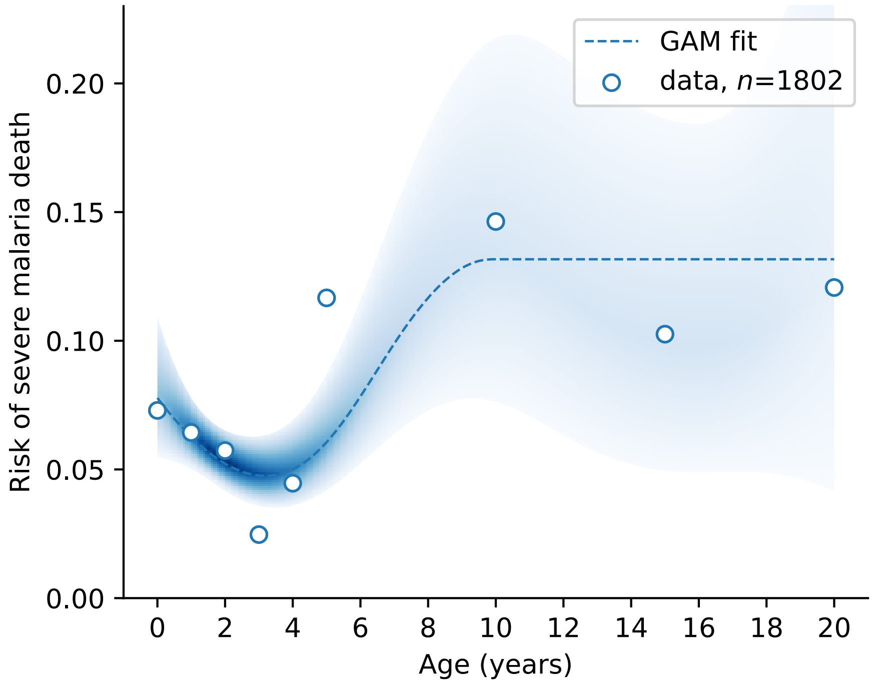

Figure 4: Severe malaria associated Estimated mean age-specific risk of death (dashed line) with 95% CO (blue band) and age-stratified data (circles) from Reyburn et al. (2005).

Bibliography

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